

Fall 2008

Problem 1 (Fa08). Find a sequence r_n of positive rational numbers such that $\sum_{n=0}^{\infty} r_n$ converges and for any prime p and any positive integer m , p^m divides the numerator of $s_k - s_j$ (written in lowest terms) for k and j sufficiently large, where $s_k = \sum_{n=0}^k r_n$.

Problem 2 (Fa08). Suppose the function f is entire, and that $f(z)/z$ is bounded in the region $|z| > 1$. Prove that $f(z) = az + b$ for some constants a and b .

Problem 3 (Fa08). Find the eigenvalues of the $n \times n$ matrix A with entries a_{ij} and $n > 2$, such that

$$a_{ij} = \begin{cases} 1 & \text{when } j - i \equiv 1 \pmod{n} \\ -1 & \text{when } j - i \equiv -1 \pmod{n} \\ 0 & \text{otherwise} \end{cases}$$

Problem 4 (Fa08). For which integer values of n is there a holomorphic function of z defined for $|z| > 1$ whose derivative is $\frac{z^n}{1+z^2}$?

Problem 5 (Fa08). Are the rings $\mathbb{R}[x]/\langle x^2 + x - 1 \rangle$ and $\mathbb{R}[x]/\langle x^2 + 2x - 3 \rangle$ isomorphic?

Problem 6 (Fa88, Fa08, Sp22). Let f be a continuous, strictly increasing function from $[0, \infty)$ onto $[0, \infty)$ and let $g = f^{-1}$. Prove that

$$\int_0^a f(x) \, dx + \int_0^b g(y) \, dy \geq ab$$

for all positive numbers a and b , and determine the condition for equality. Use this to prove Young's Inequality, which states that if p and q are positive reals with $1/p + 1/q = 1$ and a and b are positive reals then

$$\frac{a^p}{p} + \frac{b^q}{q} \geq ab$$

Problem 7 (Fa97, Fa08). Suppose N_i is a normal subgroup of a group G for $1 \leq i \leq k$, such that $N_i \cap N_j = \{e\}$ for $i \neq j$. Prove that G contains a subgroup isomorphic to $N_1 \times N_2 \times \cdots \times N_k$ if $k = 2$, but not necessarily if $k \geq 3$.

Problem 8 (Fa08). The infinitely differentiable real function $u(x, t)$ satisfies the diffusion partial differential equation $u_t = u_{xx}$ in $-\infty < x < \infty$, $t > 0$. Assume that u and all its partial derivatives of all orders are rapidly decreasing in x , in other words bounded by a constant times x^{-n} for all $n > 0$ in any strip of the form $0 < t < a$. Assume also that $\int_{-\infty}^{\infty} u(x, 1) dx = 1$. Show that if $t > 0$ then

$$\frac{d}{dt} \int_{-\infty}^{\infty} x^2 u(x, t) dx = 2$$

Problem 9 (Sp82, Fa08, Sp11, Sp25). For each of the following statements, either prove it or give a counterexample:

1. If $f_n(x)$ are continuous real valued functions on the unit interval, and f_n converges pointwise to the function f , that is, $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ for all x , then

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \int_0^1 f(x) dx$$

2. If f_n are continuous and converges uniformly to a function f then the sequence $\int_0^1 f_n(x) dx$ has limit $\int_0^1 f(x) dx$.
3. If $g(m, n)$ is real for all integers m, n , and

$$\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} g(m, n) \quad \text{and} \quad \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} g(m, n)$$

are both defined, then they are equal.

4. If the real valued functions $h_n(x)$ are continuous on the unit interval, and $\lim_{n \rightarrow \infty} h_n(x) = h(x)$ for all x , then $h(x)$ is a continuous function of x .

Problem 10 (Fa08). Show that a vector space over an infinite field cannot be the union of finitely many cosets $a_1 + V_1, \dots, a_n + V_n$ of proper subspaces $V_1, \dots, V_n$.

Problem 11 (Fa77, Su82, Fa97, Sp01, Fa03, Fa08, Sp11, Fa12, Sp18, Sp21). Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1 + x^{2n}}$, where n is a positive integer.

Problem 12 (Fa08). Let $\mathbb{F}_p$ be a field and p a prime. For $n \geq 1$ show that the number, up to isomorphism, of abelian groups of order p^n equals the number, up to similarity, of $n \times n$ matrices A over $\mathbb{F}_p$ such that $A^n = 0$.

Problem 13 (Fa08). How many complex nonreal zeros does the polynomial $z^{11} - 3z^3 + 1$ have with $1 \leq |z| \leq 2$?

Problem 14 (Fa08). Prove that the quotient of the general linear group $\mathrm{GL}_2(\mathbb{F}_3)$ by its center is isomorphic to S_4 .

Problem 15 (Fa08, Fa18). Show that $\int_0^1 \frac{1}{x^x} dx = \sum_{n=1}^{\infty} \frac{1}{n^n}$.

Problem 16 (Su77, Fa81, Sp92, Fa08). Prove that $\alpha = \sqrt{5} + \sqrt{7}$ is algebraic over $\mathbb{Q}$ by writing down a polynomial $f(x) \in \mathbb{Q}[x]$ of degree 4 having α as a zero. Show that $f(x)$ is irreducible over $\mathbb{Q}$.

Problem 17 (Fa08). We have a fair N -sided die. One side is black and the others are white. Let $n(N)$ be the smallest number of throws so the probability of getting at least one black is bigger than $1/2$. Compute

$$\lim_{N \rightarrow \infty} \frac{n(N)}{N}$$

Problem 18 (Sp80, Fa08). Let $n \geq 2$ be an integer such that $2^n + n^2$ is prime. Show that $n \equiv 3 \pmod{6}$.